

BECAMEX BUSINESS SCHOOL



PROJECT:

CONSTRUCTING A VIRTUAL INVESTMENT PORTFOLIO

Course: FIN 352 - Investments

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Table of Contents

I. Portfolio Construction.....	5
1. Investment Rationale.....	5
2. My Risk Preference & Investment Style.....	7
2.1. Investment Horizon & Risk Preference.....	7
2.2. Investment Behavior & Portfolio Style.....	7
3. Portfolio Structure.....	8
II. Risk Management Strategy.....	9
1. Capital Allocation Strategy.....	9
2. Diversification Approach.....	10
3. Risk and Reward Ratio.....	10
III. Weekly Management and Adjustment.....	11
1. Weeks 1–3: Market Analysis, Account Setup & Stock Selection.....	11
2. Week 4 - Position Building.....	14
3. Week 5 - Monitoring & Minor Adjustments.....	14
4. Week 6 - Performance Review.....	15

List of Figures

Figure1. Portforlio Structure	8
Figure 2.Sector Performance Trend	12
Figure 3.P/E for TCB, VPB, VIB	13

List of Tables

Table 1. Capital Allocation	8
Table 2.Risk - Reward Ratio	10
Table 3.Changes in Stocks	14

I. Portfolio Construction

1. Investment Rationale

In 2025, Vietnam's economy is considered to be in an expansion phase, with positive signals from GDP growth, public investment, domestic consumption, and digitalization policy. According to the National Statistics Office of Vietnam, GDP in the first six months of 2025 increased by 7.52% over the same period last year, the highest level of the first six months in the period 2011-2025. Vietnam prioritizes the development of strategic infrastructure, particularly national key transport systems, including expressways, high-speed railways, inter-regional projects, large-scale urban infrastructure, and digital transformation infrastructure (ThuVienPhapLuat, 2025). This creates growth opportunities for sectors such as construction materials, industrial real estate, and companies linked to infrastructure supply chains. The government has introduced various incentive policies to encourage enterprises—especially small and medium-sized enterprises (SMEs)—to invest in digital transformation, research and development (R&D), and technological innovation during the 2025–2030 period (Government e-Newspaper, 2025). The trend toward cashless payments, along with regulations requiring businesses and stores to use bank accounts, provides fertile ground for digital banks to grow. On the energy front, Thi Hà and Phuong Dung (2025) reported that since August 2025, Vietnam began piloting E10 fuel in three major cities, aiming for nationwide implementation in early 2026. This policy acts as a catalyst for oil refiners and biofuel producers. Overall, Vietnam's macroeconomic conditions are favorable for the stock market, especially in sectors that benefit directly from public investment, digital transformation, and new energy policies.

The basic materials sector—particularly construction steel—directly benefits from public investment and key infrastructure projects such as expressways, high-speed railways, and large-scale urban development. The three leading steel companies in the basic materials

sector are Hoa Phat (HPG), Hoa Sen (HSG), and Nam Kim (NKG)—each holding significant market share and large-scale production capacity in Vietnam. Among these, I selected Hoa Phat (HPG) due to its dominant market position, strong fundamentals, and direct exposure to the country's infrastructure expansion strategy.

Next, the government promotes digital transformation in the financial sector through policies that encourage cashless payments. Businesses and retailers are now required to use bank accounts for transactions, and lead to significant demand for digital banking services. Potential candidates in the banking sector include Techcombank (TCB), VPBank (VPB), and Vietnam International Bank (VIB). While VIB focuses on retail and consumer finance, and MBB shows strong growth in military and public sector segments, TCB stands out for its leading digital banking platform, high CASA ratio, and efficient cost structure. With a strategic focus on affluent customers and strong return metrics, TCB was selected as the representative for the financial sector in my portfolio.

Finally, E10 is a biofuel blended with 10% ethanol and 90% traditional gasoline, requiring advanced refining capacity to meet environmental and quality standards. The oil refining and petrochemical industry benefits directly from this policy, especially companies with large-scale operations and domestic supply capabilities. Potential candidates in the energy sector include Binh Son Refining and Petrochemical (BSR), Petrolimex (PLX), and PVOil (OIL). While PLX and OIL primarily operate in fuel distribution, BSR stands out for its large-scale refining capacity and its direct alignment with the government's E10 fuel policy. BSR is the most suitable choice for my portfolio.

My portfolio consists of three stocks from three distinct sectors, each directly benefiting from national policy priorities, thereby ensuring both short-term upside potential and long-term sustainability. The portfolio structure also captures key macro trends—public

infrastructure investment, digital transformation, and clean energy—providing a solid foundation for investment performance.

2. My Risk Preference & Investment Style

2.1. Investment Horizon & Risk Preference

I intend to invest over a medium-to-long-term horizon of 5 to 7 years, with the primary goal of achieving capital growth while protecting a portion of my initial investment. Bodie et al. (2018) explain that investors with longer timeframes can accept a moderate level of risk in exchange for higher expected returns over time. Therefore, I adopt a strategic asset allocation approach—balancing risky and risk-free assets at the outset—to align my portfolio structure with both my investment goals and time horizon. This strategy also reflects an understanding of the risk-return trade-off, in which I am willing to accept a certain level of risk to achieve long-term capital growth, while avoiding excessive losses. In my case, I can tolerate a maximum loss of around 20% on a single investment, which indicates that I am moderately risk-averse.

2.2. Investment Behavior & Portfolio Style

As I do not follow the market daily and review my portfolio only a few times a month, I prefer a balanced portfolio structure that requires infrequent adjustments. However, when I identify a company with strong potential, I am willing to spend more time researching that firm and the market conditions to make informed decisions. Overall, I seek to maintain a mix of safe assets and medium-risk assets to match both my risk tolerance and time commitment to investing.

3. Portfolio Structure

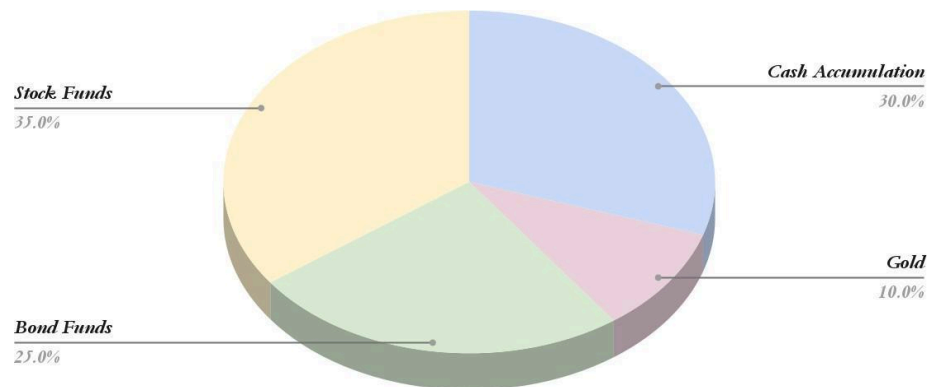


Figure1. Portfolio Structure

My portfolio includes four asset classes: cash accumulation (30%), gold (10%), bond funds (25%), and equity funds (35%). This allocation reflects my moderate risk tolerance and long-term investment horizon of 5–7 years. I aim to grow my capital over time while managing risk through careful asset selection. According to strategic asset allocation theory, setting target weights for each asset class helps align the portfolio with both return objectives and risk capacity. To reduce overall portfolio risk, I apply the principle of diversification, which involves holding a mix of assets that respond differently to market conditions. For example, while equity funds offer growth potential, bond funds and gold provide stability and act as hedges during periods of volatility. By combining risky and safe assets, my portfolio benefits from the diversification effect, which lowers total risk without sacrificing expected return.

II. Risk Management Strategy

1. Capital Allocation Strategy

Table 1. Capital Allocation

Asset Class	Allocation (%)	Amount (VND)
Cash / Money Market Instruments	30%	150,000,000
Gold	10%	50,000,000
Bond Fund	25%	125,000,000
Equity	35%	175,000,000

In accordance with the principles of Asset Allocation Theory, portfolio returns are largely determined by the strategic distribution of capital among different asset classes, rather than by individual security selection or market timing (Elton et al., 2014). My portfolio is structured to reflect a moderately risk-neutral profile, aiming for long-term capital growth while maintaining sufficient liquidity and stability. With a total investment capital of 500 million VND .The strategic purpose of this allocation is to balance growth potential with capital preservation by combining assets with differing risk-return profiles and low correlations. Cash holdings (30%) provide high liquidity, allowing for rapid reallocation in response to market opportunities or downturns. Gold (10%) serves as a hedge against inflation, currency depreciation, and systemic market shocks, adding stability in times of macroeconomic uncertainty. The bond fund allocation (25%) offers predictable returns and low volatility, helping to stabilize overall portfolio performance. Finally, equities (35%) function as the portfolio's primary growth engine, targeting companies with strong fundamentals and favorable sectoral trends.

2. Diversification Approach

Diversification is a key principle in portfolio construction, aiming to reduce unsystematic risk by spreading investments across assets that are not perfectly correlated (Markowitz, 1952). When the returns of assets move independently from one another, the volatility of the overall portfolio (σ_p) can be reduced even if individual assets are volatile. In this portfolio, diversification is achieved by selecting stocks from three distinct sectors: basic materials, banking, and energy. The rationale behind this sectoral mix is that macroeconomic or industry-specific shocks are less likely to affect all sectors simultaneously. For example, a downturn in the steel industry due to commodity price fluctuations may have little to no direct impact on banking profitability or energy demand. This cross-sector allocation reduces exposure to sector-specific risks, contributing to more stable portfolio performance.

3. Risk and Reward Ratio

I allocated 35% of my total portfolio value (500 million VND)—equivalent to 175 million VND—to stocks, reflecting my balanced risk preference and medium-to-long-term investment horizon. My maximum loss tolerance for the stock allocation is 20%, which equals 35 million VND. Based on this, I structured my positions with a clear priority order: HPG as the highest conviction pick, followed by TCB and then BSR.

The table below illustrates the capital allocation, number of shares, stop-loss and target levels, and the calculated Risk–Reward (R:R) Ratios for each stock. HPG, with the highest R:R ratio of 4:1, receives the largest allocation at nearly 70 million VND, reflecting both its strong upside potential and my confidence in its fundamentals. TCB, with an R:R of 2.33:1, is allocated over 61 million VND due to its stable growth prospects in the banking sector. BSR, while still offering a positive R:R of 2:1, has the smallest allocation, acknowledging its relatively higher exposure to commodity price volatility. This structured approach ensures that capital is distributed in alignment with both my risk tolerance and

expected returns, while maintaining portfolio diversification across three distinct sectors to mitigate sector-specific risks.

Table 2. Risk - Reward Ratio

	HPG	TCB	BRS
Entry Price	25,500	35,050	19,800
Number of Shares	2,745	1,748	2,209
Capital Allocation	69,997,500	61,267,400	43,738,200
Stoploss (%)	25%	15%	15%
Stop-loss Price	19,125	29,793	16,830
Risk per share	6,375	5,258	2,970
Target (%)	100%	35%	30%
Target Price	51,000	47,318	25,740
Reward per share	25,502	12,268	5,940
R:R Ratio	1:4	1:2.33	1:2

III. Weekly Management and Adjustment

1. Weeks 1–3: Market Analysis, Account Setup & Stock Selection

In my first week, I focused on understanding the overall market rather than making any trades. I tracked the VN-Index and sector performance while reviewing Vietnam’s economic outlook and key government policies. The economy showed strong momentum,

with GDP in the first half of 2025 growing by 7.52%—the highest since 2011. The main drivers included accelerated public investment, especially in infrastructure, and robust domestic consumption supported by post-pandemic recovery and improvements in global trade flows. The government’s push for strategic infrastructure development, a nationwide cashless payment mandate, and the E10 biofuel rollout created distinct growth opportunities for my three target sectors.



Figure 2. Sector Performance Trend

In Week 2, I officially registered for the Vietstock Investment Challenge and created my simulation trading account. This step marked my transition from theoretical analysis to practical portfolio preparation. The platform provided real-time market data and execution features, allowing me to simulate trading under realistic conditions without risking real capital. I familiarized myself with the trading interface, order types, and portfolio tracking tools, ensuring I could execute trades efficiently in later weeks.

For banking, I compared Techcombank (TCB), VPBank (VPB), and VIB using historical P/E ratios from 2020 to 2024. TCB consistently maintained a relatively low and

stable P/E range (3–6), suggesting a balance between valuation and earnings stability, while VPB’s low P/E reflected higher perceived risk, and VIB’s more volatile multiples suggested less earnings predictability. Coupled with TCB’s leading digital banking platform and high CASA ratio, it became the optimal choice to capture upside from the government’s cashless payment mandate.

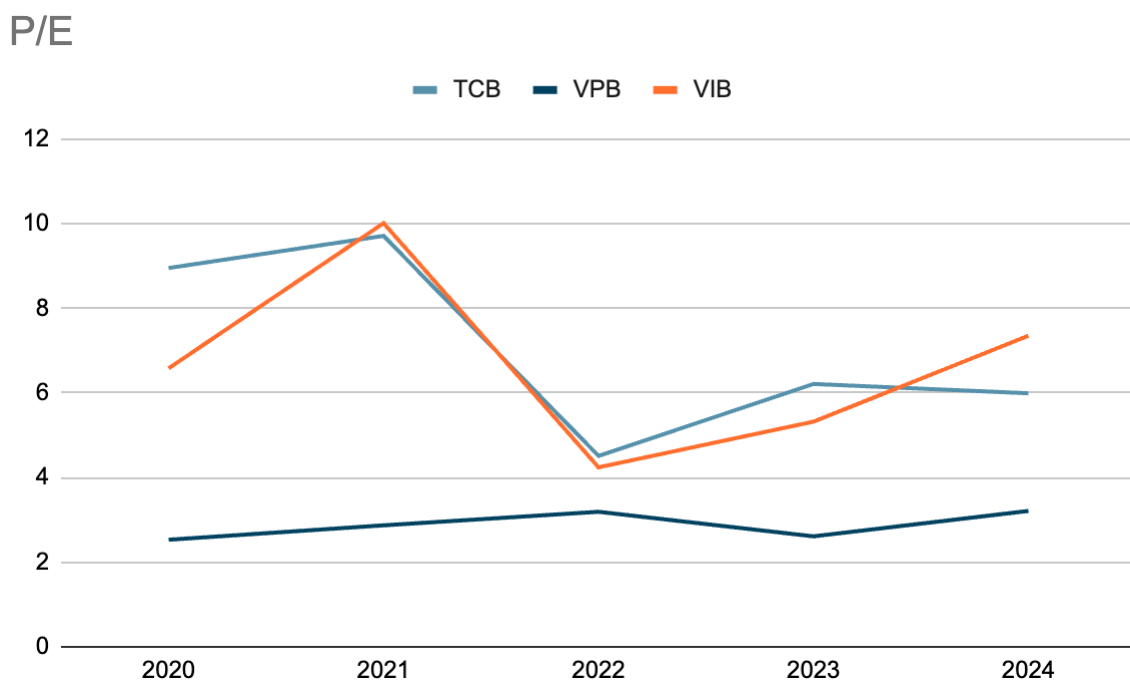


Figure 3.P/E for TCB, VPB, VIB

In energy, I shortlisted Binh Son Refining and Petrochemical (BSR), Petrolimex (PLX), and PVOil (OIL). PLX and OIL focus on distribution, whereas BSR is the sole domestic refiner capable of producing E10 biofuel—aligning directly with the government’s rollout policy. BSR’s role as a supplier to PLX and OIL further reinforced its strategic importance, making it the preferred selection for my portfolio. While HSG relies heavily on distributing products purchased from HPG and NKG lacked sufficient public data, HPG

stood out as both a domestic and export-oriented steel producer with significant capacity to supply national infrastructure projects.

By the end of Week 3, I had finalized my initial portfolio: HPG for infrastructure-driven steel demand, TCB for digital banking growth, and BSR for energy policy alignment. These selections reflected both macroeconomic tailwinds and company-specific strengths, forming a diversified yet focused foundation for the next phases of the competition.

2. Week 4 - Position Building

I initiated my first trades in Week 2 to Week 4. I allocated 35% of my total capital (500 million VND)—equivalent to 175 million VND—to equities, in line with my capital allocation strategy defined in the Risk Management section. This equity portion was distributed across three selected stocks: HPG (steel, basic materials), TCB (banking), and BSR (energy). My pre-set risk tolerance of a maximum 20% loss on the equity portion. I executed buy orders at the following entry prices: HPG at 25,500 VND, TCB at 35,050 VND, and BSR at 19,800 VND, with respective capital allocations of 69,997,500 VND (40% of equity portion), 61,267,400 VND (35%), and 43,738,200 VND (25%).

3. Week 5 - Monitoring & Minor Adjustments

In Week 5, In Week 5, I focused on monitoring my three equity positions—HPG, TCB, and BSR—by tracking both market price movements and sector-specific news. HPG advanced about 3% from its entry price, supported by announcements of accelerated approvals for public infrastructure projects. Since the price remained far from my target level of VND 51,000 and comfortably above the stop-loss of VND 19,125, I decided to hold the position without making adjustments.

TCB traded within a narrow range, decreasing roughly -3.3% from its entry price. The price action did not approach either my target or stop-loss levels, so I maintained the position unchanged. BSR showed slightly more volatility, moving within a $\pm 2\%$ range, influenced by fluctuations in global crude oil prices. However, domestic energy news reaffirmed the government's plan to roll out E10 fuel nationwide in early 2026, strengthening its long-term growth outlook. Based on this positive policy momentum, I revised my target price for BSR upward by 5%, while keeping the stop-loss level intact to preserve my original risk-reward balance.

Table 3.Changes in Stocks

Stock	Entry Price	Current Price	% Change	Action
HPG	25,500	26,265	+3.00%	Hold
TCB	35,050	33,900	-3.28%	Hold
BSR	19,800	20,196	$\pm 2.00\%$	Hold + Increase Target

4. Week 6 - Performance Review

As of August 8, 2025, my portfolio showed solid performance across all three holdings. HPG rose from an entry price of 25,500 VND to 28,000 VND (+9.80%), driven by strong Q2 earnings and sustained demand from Vietnam's infrastructure expansion. TCB increased from 35,050 VND to 37,850 VND (+8.00%), supported by robust credit growth data and positive momentum in digital banking adoption. BSR recorded the highest gain, moving from 19,800 VND to 22,200 VND (+12.12%), as higher oil prices and the nationwide E10 fuel rollout boosted investor sentiment.

However, while reviewing valuation metrics such as the P/E ratio, I realized that my earlier comparisons between companies were partially based on irrational investor behavior—focusing on relative multiples without fully considering the differences in business models, growth prospects, and sector risks. I also attempted to calculate expected returns using the CAPM model in Excel, but the results were inconsistent due to data and beta estimation challenges. This highlighted the importance of accurate input data and understanding the assumptions behind financial models.

From a personal development perspective, this project has been transformative. Coming from a non-finance background, I initially approached the competition with limited technical knowledge. Over six weeks, I became familiar with core concepts in finance, including risk-return trade-offs, diversification benefits, and practical stock selection.

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